INF221 - [Modern] Web Design & Development

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Introduction to web development processes

Key Processes

- ❖ Understanding the business process
  - ➢ Defining the web project(s) goal(s)
- ❖ Web development team

Web development phases

Models & Methodologies

Project Management - Composing project teams

- Project success is based on choosing the right people for the project team and obtaining the necessary level of commitment from them
- The work breakdown structure (WBS) can be used to identify the work to be done
  - after which you will have a reasonable idea of the work activities that need to be carried out,
  - the number of staff needed to undertake the work
  - and the specialist skills and knowledge required
- The list below can be used as a guide to putting together a balanced team:
  - A single coordinator or shaper (but not both) as team leader
  - A plant to generate ideas
  - A monitor-evaluator to provide objective analyses
  - One or more:
    - implementer
    - team worker
    - resource investigator
    - completer-finisher

Defining Project Goals & Scope

- ❖ Planning a project is about deciding what work needs to be done in order to achieve the objectives
  - ➢ who will undertake each task involved
  - ➢ when each task will be carried out and how long the work will take
  - ➢ and what resources will be required.
- ❖ In addition, all of these activities must be viewed against the backdrop of cost
  - ➢ both ongoing and final.
- ❖ The plan addresses questions such as ***what***, ***when***, ***how*** and ***by whom***
- ❖ User requirements define the project deliverables
  - ➢ which determines what needs to be developed
- ❖ Project scope defines project boundaries
  - ➢ determine both the extent of the resources required
  - ➢ and the amount of time needed to complete the project

Key Processes

Key Processes/Activities:

- ❖ Planning and Design, which create the appearance, layout and style that users see
- ❖ Development, which brings this imagery to life as a functioning web tool

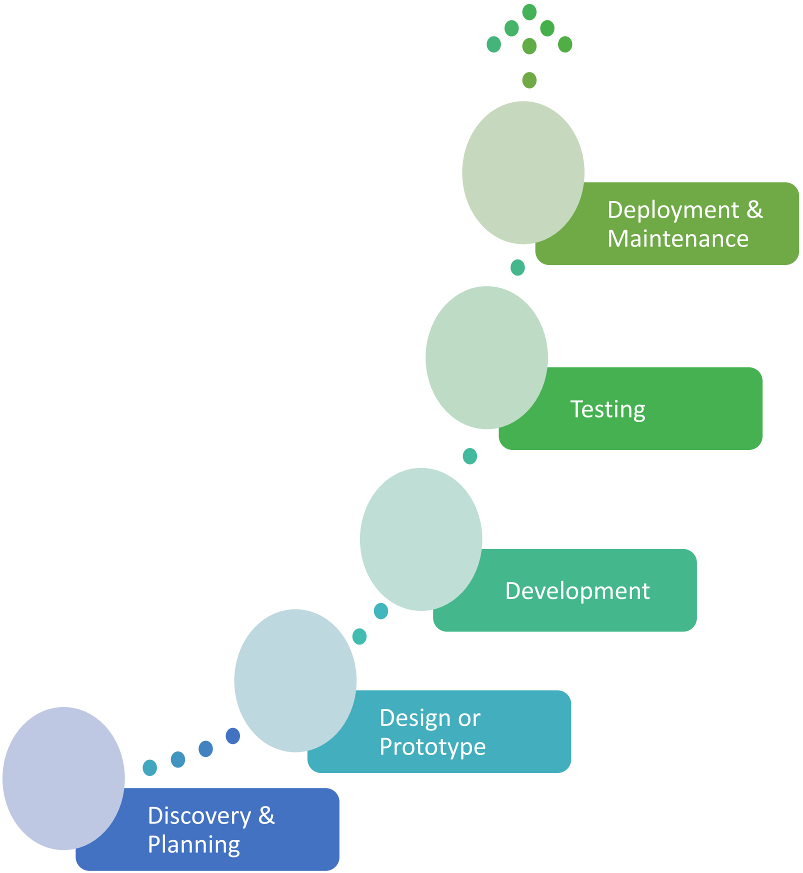
Iteration is key throughout the Core activities

To manage change and risk of failure, the systems life cycle breaks a project down into a series of well-defined stages

- ❖ from the formulation of the initial concept through to the implementation and maintenance of a fully working system

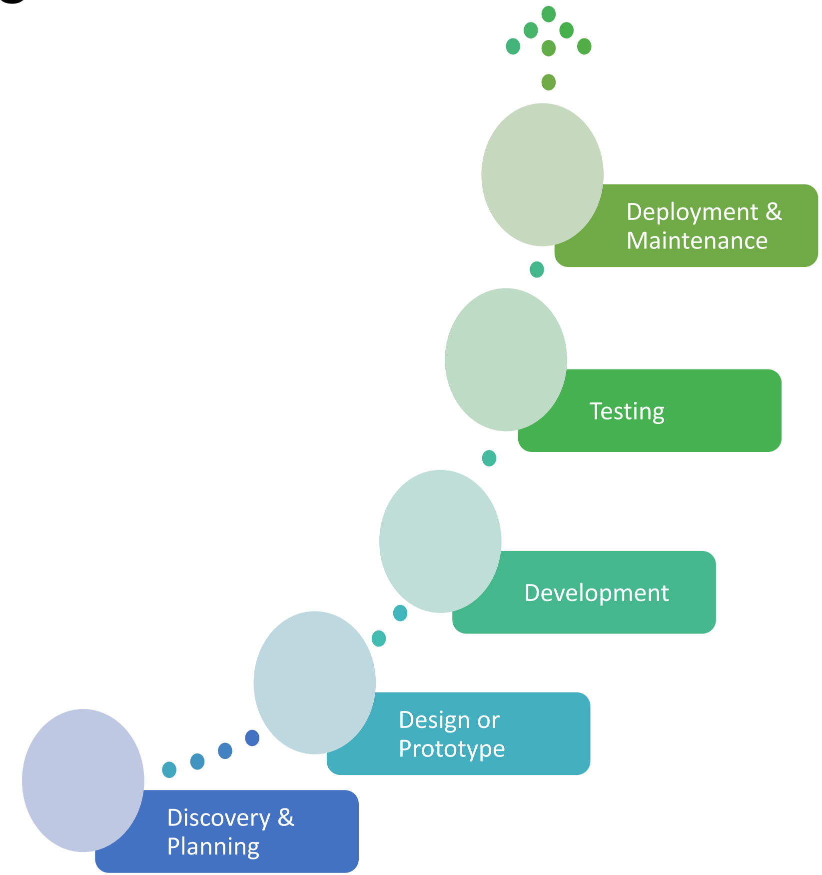
Project Life Cycle Stages

- Requirements analysis
  - the scope of the system is defined, and a set of requirements is derived
    - (i.e. the functionality and levels of service that the new system must provide)
- Design
  - designed to meet the requirements specified as the result of the previous phase
- Development
  - system built.
  - hardware is installed, database and software applications are written, and data is loaded into the system
- Testing
  - depending on the model in use, testing may be done during, or after actual development & deployment of the system software
- Maintenance
  - once the client has accepted the new system and it is in daily use
  - there will inevitably be adjustments to be made to the system throughout its useful life
  - hardware elements will need to be replaced or upgraded
  - and the need for additional functionality
  - or information processing features may necessitate modifications to software or database programs



Web Development Processes

- Discovery & Planning – Requirements gathering
- Users must be involved – they are critical
- Input - Reports from clients and documentations from Business Analyst
- Output - Complete final project documentation with requirement specifications for designers as well as developers and Wireframes



Web Dev. Processes - Business & Users

- The key principle to good design & development is:
  - To understand your users - they are the people who will actually be using and interacting with your web application.
    - Who are they?
    - What activities are being carried out?
    - What are they looking for?
    - What are their objectives?
- User Experience (UX) must be central to the process.
  - Need to optimize the interactions users have with a product:
    - So that they match the users' activities and needs
  - Understanding users' needs
    - Take into account what people are good and bad at
    - Consider what might help people in the way they currently do things
    - Think through what might provide quality user experiences
    - Listen to what people want and get them involved
    - Use tried and tested user-centered methods

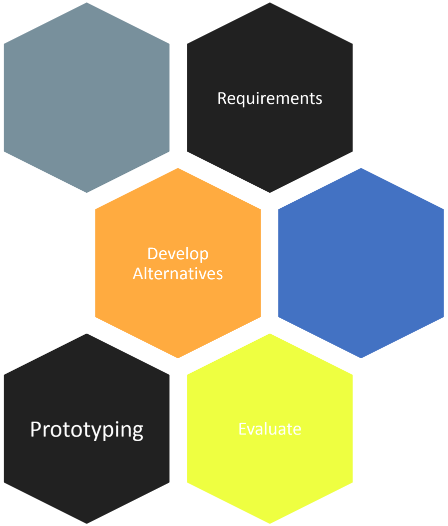
Web Dev. Processes

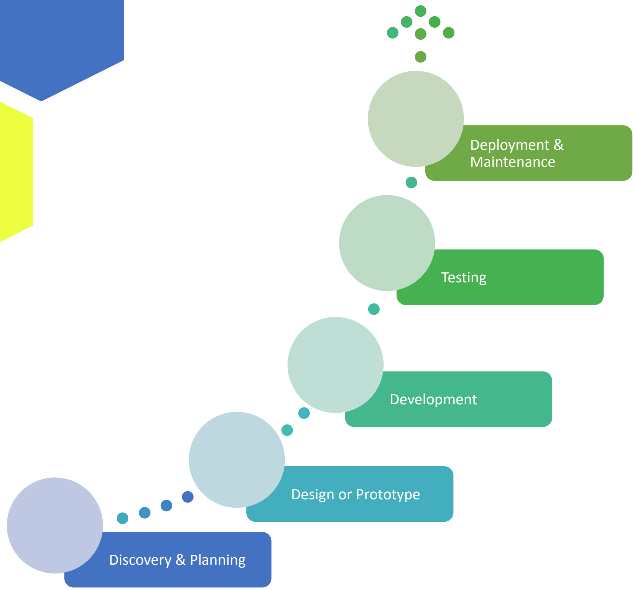
Design or Prototype

Understanding & Conceptualizing Design

Steps involved in the Design

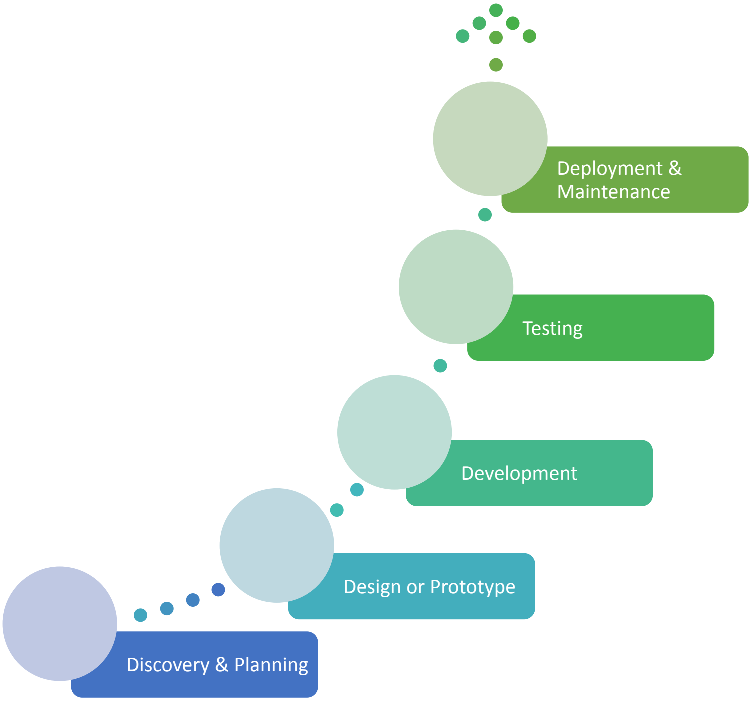
- ❖ Establishing requirements
- ❖ Developing alternatives
- ❖ Prototyping
- ❖ Evaluating





Web Dev. Processes

❖ Development

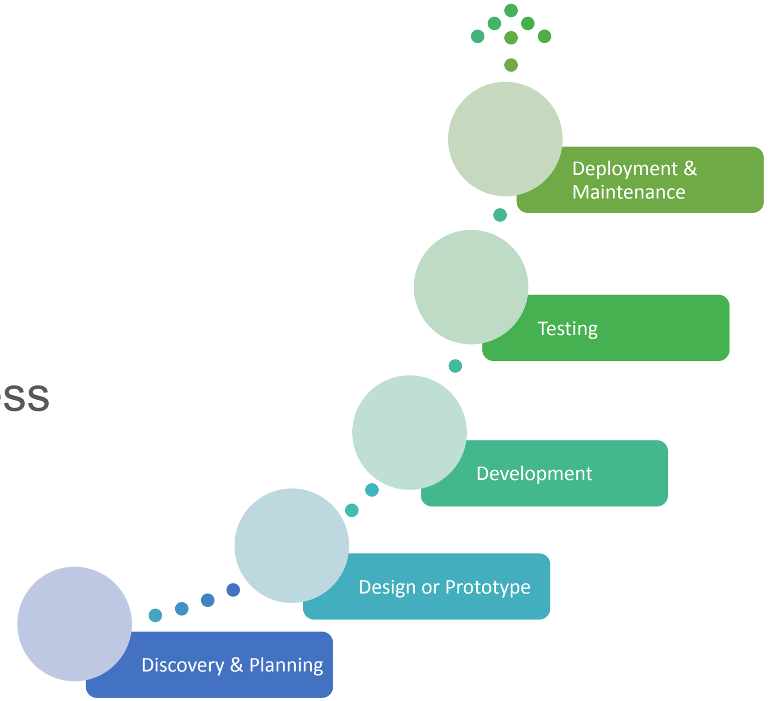


Web Dev. Processes

Testing

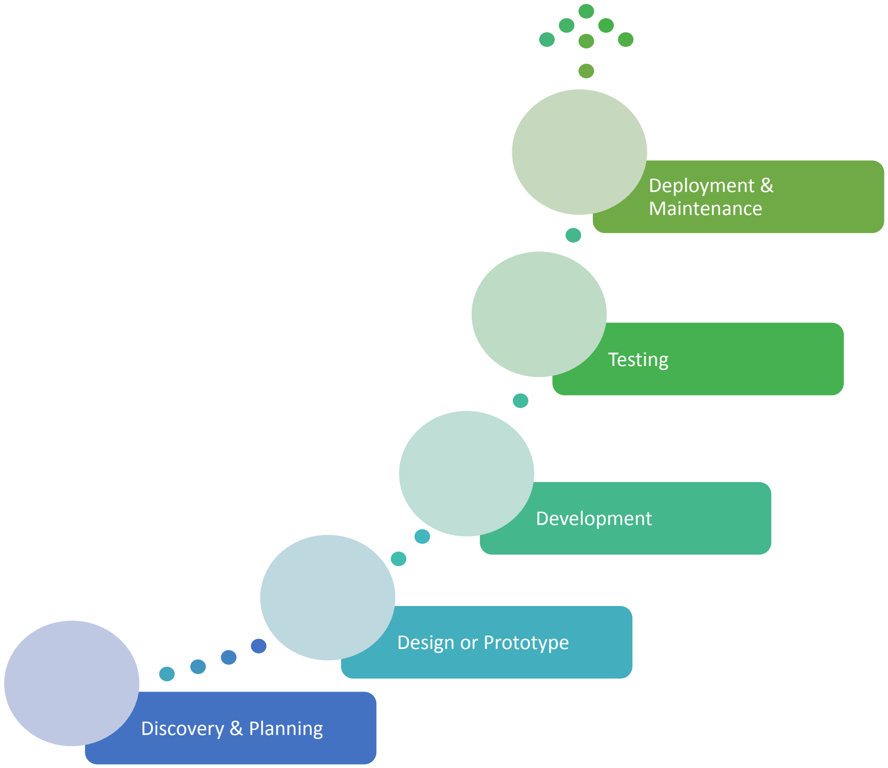
- ❖ Unit Testing
- ❖ UI Testing
- ❖ Integration Testing
- ❖ E2E

Iterative manner throughout the process



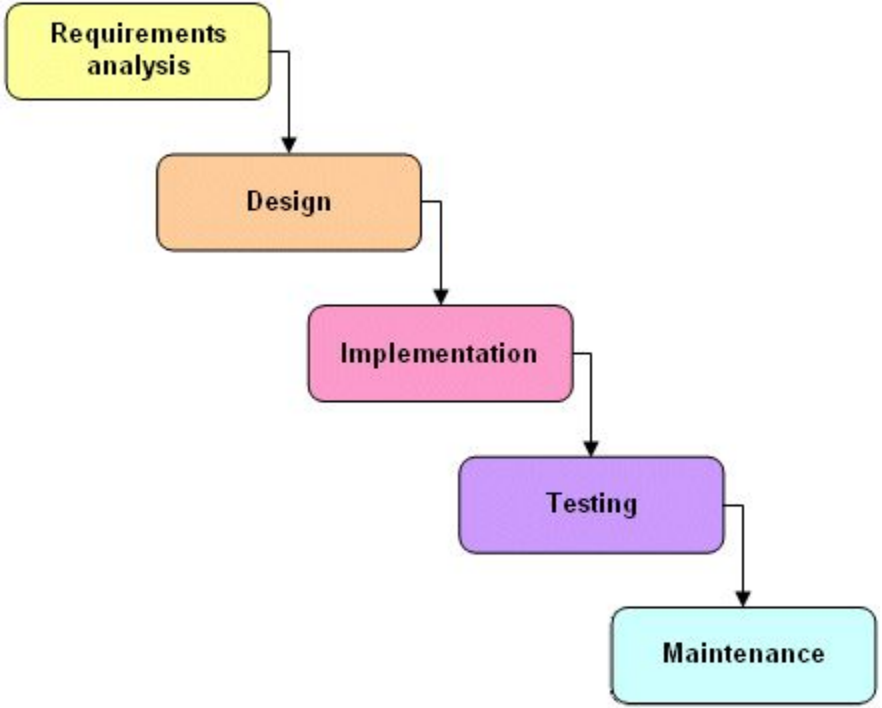
Web Dev. Processes

- Deployment – production
- Finished product
  - enables users to achieve their daily work
- Maintenance – mostly forgotten activity
  - especially in most projects at home – Malawi.



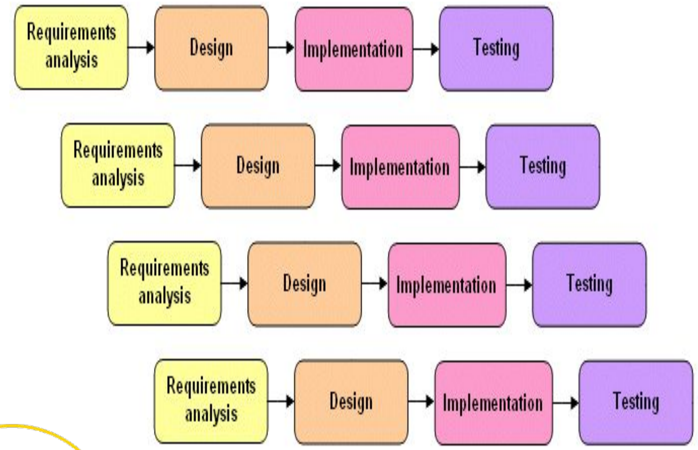
Development Lifecycle Models - Waterfall

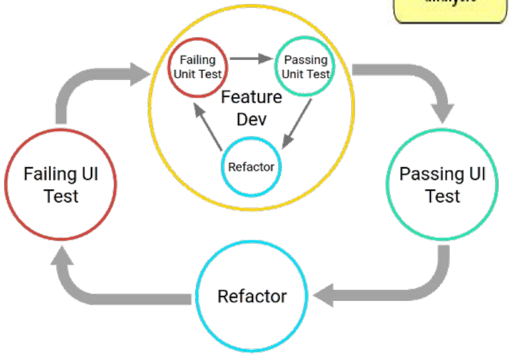
- move to the next phase only when the current phase is complete
- proceed from one phase to the next in a purely sequential manner
- phases of development are discrete, and do not allow for jumping back and forth, or any overlap between the phases
  - although there are modified versions of the waterfall model that are not so rigid



Development Lifecycle Models - Incremental Model

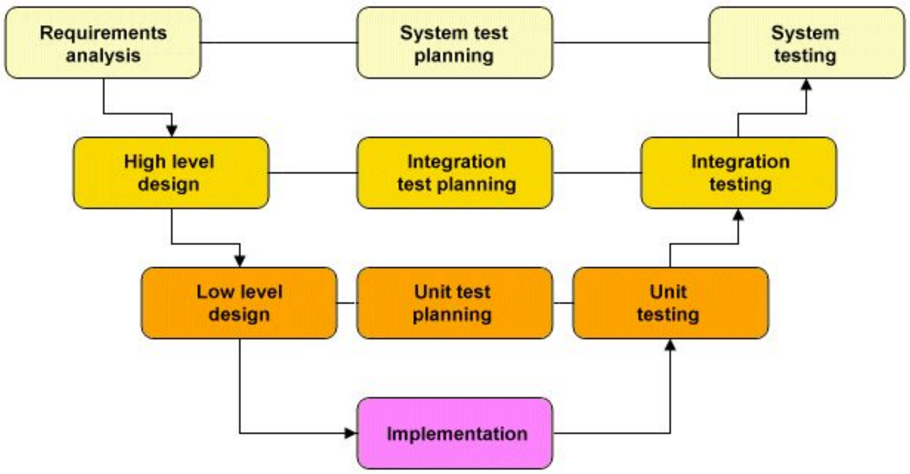
- ❖ Allows different parts of the system to be developed separately, each with its own life cycle iteration
- ❖ Each iteration includes a requirements analysis, design, implementation and testing phase
  - ➢ The first iteration usually produces a working version of the system on which subsequent iterations build.
- ❖ The incremental model provides a more flexible and less costly alternative to the waterfall model
  - ➢ in which changes to the scope or requirements of the project can be more easily accommodated.
- ❖ Each iteration can be completed relatively quickly, and is easier to manage.
- ❖ Like the waterfall model, however, each phase of each iteration must be completed before the next phase can commence,
  - ➢ and the overall system architecture may be poorly defined in the early stages of the project and evolve it as project progresses





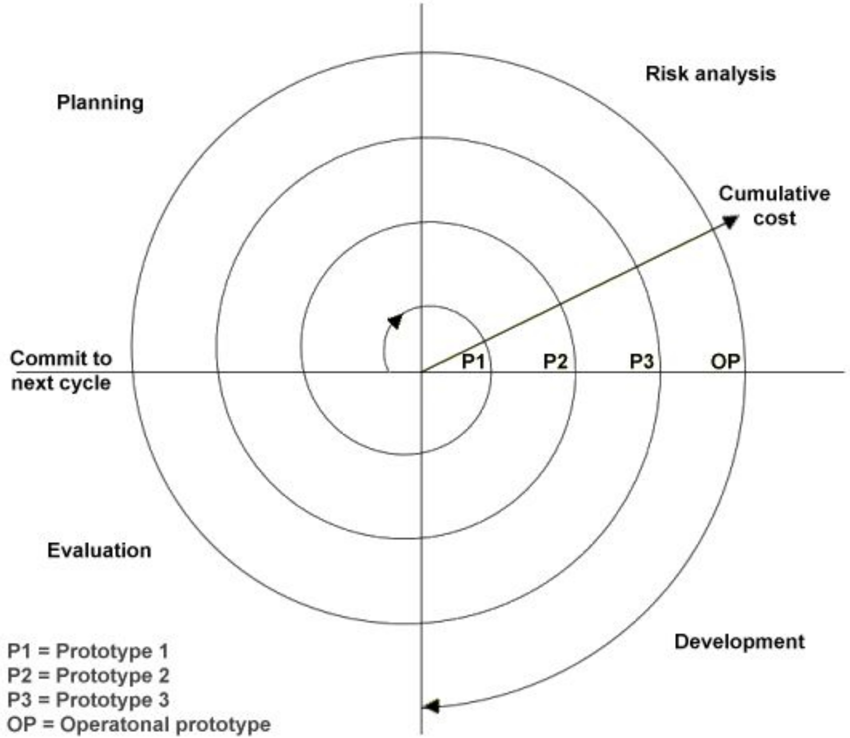
Development Lifecycle Models - The V-shaped model

- Like the waterfall model, each phase of the V-Shaped life cycle must be completed before the next phase begins
- Much more emphasis on testing
  - Test procedures are developed early in the life cycle
    - unit testing is planned to assess the success of the low-level design phase, which concentrates on individual system components
    - integration testing is developed during the high-level design phase, which focuses on the overall system architecture
    - system testing is designed to assess whether the requirements determined during the requirements analysis phase have been met



Development Lifecycle Models - The Spiral model

- The spiral model is similar to the incremental model
  - with more emphasis on risk analysis
- Project passes repeatedly through the four phases of planning, risk analysis, development and evaluation
- Cost of the project increases as the spiral goes through successive iterations
  - high degree of risk analysis and the production of a prototype system early in the life cycle makes this a good option for large and complex projects
    - although it can be potentially costly, and
    - the level of expertise required for the critical risk analysis phase means that it is not really suited to smaller projects



Development Methodologies

Describe a framework for the development of information systems

- A particular methodology is usually associated with a specific set of tools, models and methods
  - for the analysis, design and implementation of information systems
  - and each methodology tends to favour a particular lifecycle model

Development Methodologies - Agile Methodologies

- Refers to a group of loosely related software development methodologies that are based on similar principles
- Each iteration of the project life cycle results in the production of working software, which is then evaluated by the customer before the next iteration begins
- Production of working software, rather than the completion of extensive project documentation, is seen as the primary measure of progress
  - The software produced at the end of an iteration has been fully developed and tested, but embodies only a subset of the functionality planned for the project as a whole.
    - The aim is to deliver functionality incrementally as the project progresses.
    - Further functionality will be added, and existing features will be refined, throughout the life of the project

Question Time

- ❖ List and explain any advantages and disadvantages of each model discussed above
- ❖ List & explain any advantages and disadvantages of each methodologies discussed above

Summary

Key Processes

- ❖ Understanding the business process
    - ➢ Defining the web project(s) goal(s)
- ❖ Web development team
- ❖ Web development phases

Development Models & Methodologies

- ❖ Waterfall
- ❖ Agile

References

- http://en.wikipedia.org/wiki/List_of_software_development_philosophies
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